SUBMISSION HIGHLIGHTS: PAPER II (v3.0)

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• Spacetime Algebra $Cl_{1,3}$ multivector rotors eliminate quench singularities and enforce real energy density positivity ($T_{00} \geq 0$).

• Turn-to-turn contact resistivity of 15.0 μΩ·cm² enables rapid radial current shunting during localized quenches.

• Passive current decay bounds peak quench hot-spot temperatures at $T_{max} \leq 200$ K, protecting conductor matrix within its reversible elastic strain window ($-0.30\% \leq e_{ax} \leq +0.35\%$).